BRITISH INTERNATIONAL

SCHOOL OF SULAYMANIYAH

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Information & Communication Technology

OVERVIEW

- This chapter covers a wide range of ICT uses in daily life and industry.
- Key areas include communication, modelling, and computer control.
- We will explore financial, medical, and retail applications.
- The role of satellite and recognition systems is examined.
- The content connects with previous knowledge on hardware and software.
- Examples provided are relevant to real-world scenarios.

6.1

Communication

Modern Communication Methods

- ❖ ICT technology drives various communication systems.
- ❖ Common methods include newsletters, posters, and websites.
- ❖ Multimedia presentations and media streaming are widely used.
- ❖ E-publications are replacing traditional paper formats.
- ❖ Mobile communication has revolutionized how we connect.
- ❖ Each method has specific use cases and target audiences.

Production & Use

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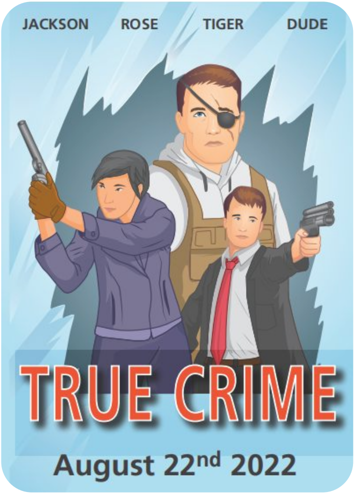
Guidelines for Newsletters
Avoid squeezing too much information onto a single page.
Use clear, easy-to-read fonts like Arial or Trebuchet MS.
Font sizes of 11, 12, or 14 points are recommended.
Use columns to organize text and images effectively.
Avoid excessive capital letters; use bold for headings instead.
Real photos are preferred over clip art for engagement.

Posters for Events

Essential Information

- ❖ Sporting event posters must include the event type and location.
- ❖ Date, time, and admission fees are critical details.
- ❖ Contact details and facility information (e.g., parking) are needed.
- ❖ Movie posters require a representative image of the genre.
- ❖ Release dates and main character lists are standard.
- ❖ Strategic placement allows targeting of specific audiences.



Websites as opposed to print media

- ❖ Websites offer worldwide advertising capability.
- ❖ No need for paper, consumables, or physical distribution.
- ❖ Companies can develop their own or pay for ad space.
- ❖ Requires investment in design, hardware, and maintenance.
- ❖ Security against hackers and pharming is a necessity.
- ❖ Content management systems make self-hosting easier.

Advantages of Websites

- ❖ Sound, video, and animation can be integrated.
- ❖ Hyperlinks allow easy navigation to related pages.
- ❖ Hit counters provide data on visitor numbers.
- ❖ Global reach extends beyond local markets.
- ❖ Updates are easier and faster than reprinting paper.
- ❖ Content cannot be physically defaced or discarded.

Disadvantages of Websites

- ❖ Risk of hacking, modification, or virus introduction.
- ❖ Pharming attacks can redirect users to fake sites.
- ❖ Customers require internet access and a computer/device.
- ❖ Less portable than paper (though smartphones are changing this).
- ❖ Possible for customers to go to undesirable websites
- ❖ Maintenance costs can be high for professional sites.
- ❖ Harder to target a specific local audience compared to posters.
- ❖ The need to find a way for people to find out about the website

Advantages

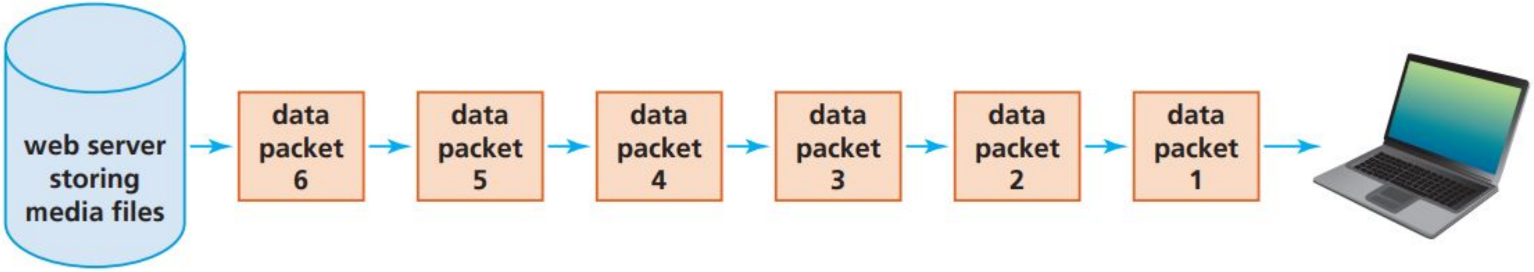
- ❖ Uses animation, video, sound, and music for interest.
- ❖ Produced with software and displayed via projectors.
- ❖ Sound and video grab audience attention effectively.
- ❖ Interactive hyperlinks can access websites or cloud files.
- ❖ Transition effects display facts in chronological order.
- ❖ Presentations can be tailored to different audiences easily.

Disadvantages

- ❖ Specialized equipment can be expensive to procure.
- ❖ Risk of equipment failure during the presentation.
- ❖ Internet access may be required for certain features.
- ❖ Focus can shift to the medium rather than the message.
- ❖ Overuse of effects can result in a "bad" presentation.
- ❖ Too much text or animation can distract the audience.

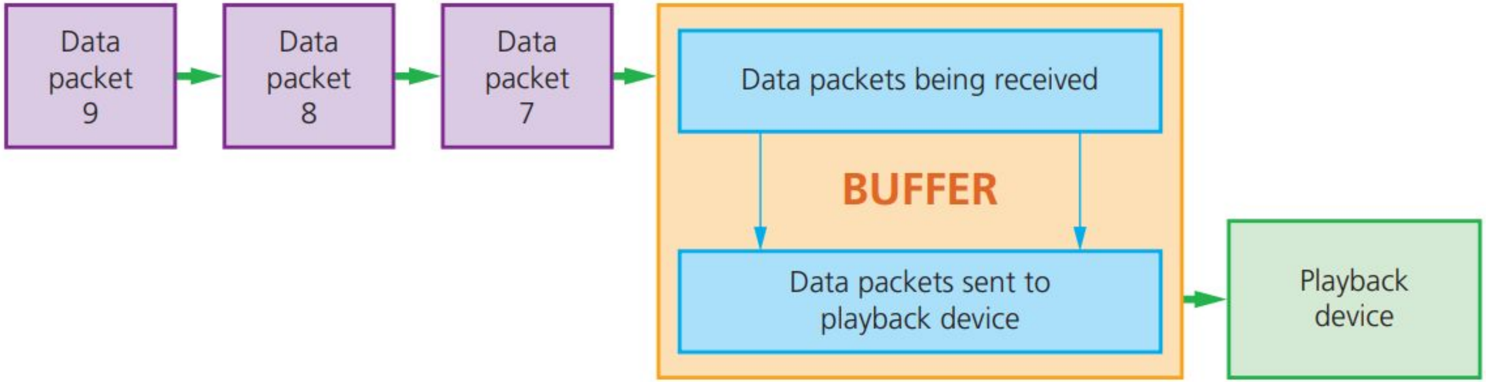
Media Streaming

- ❖ Users watch or listen without downloading the full file.
- ❖ Continuous transmission from a remote server to the device.
- ❖ Data is transmitted and played in real-time.
- ❖ Saves local storage space on HDD or SSD.
- ❖ Avoids long wait times associated with full downloads.
- ❖ Requires a stable internet connection (e.g., 25 Mbps for HD).



How Streaming Works
Buffering and Playback

- ❖ Files are sent as a series of data packets.
- ❖ Data packets are buffered (stored temporarily) in the computer.
- ❖ Buffering ensures smooth playback without freezing.
- ❖ The buffer sends data to the playback device continuously.
- ❖ Large buffers reduce the risk of playback gaps.

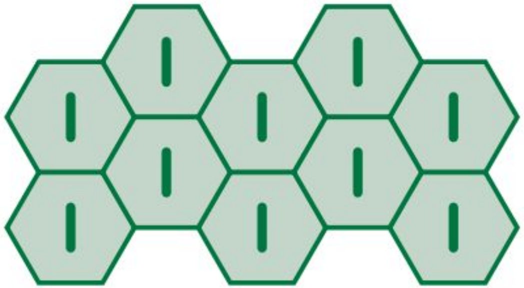


E-publications

- Includes e-books, digital magazines, and newspapers.
- Downloaded to devices for reading via internet connection.
- Navigation is done by swiping the screen.
- Text size can be expanded for better readability.
- Multimedia elements can be embedded in the text.
- Cheaper to produce as there are no printing costs.

Cellular Networks

- ❖ Mobile phones use a network of towers inside cells.
- ❖ Towers allow data transmission across large areas.
- ❖ Signals weaken at cell edges and switch to adjacent towers.
- ❖ Satellite technology links calls between different countries.
- ❖ Devices use SIM cards or wireless internet to connect.
- ❖ Supports SMS, calls, VoIP, and internet access.



Cell showing tower at the centre. Each cell overlaps giving mobile phone coverage

Short Message Service

- Quick communication via typed text on mobile devices.
- Recipients do not need to be immediately available.
- Often free between phones on the same network.
- Predictive texting speeds up input by completing words.
- Useful for communicating when phone calls are not possible.
- Can be sent even if the recipient's phone is off.

Voice Communication

- ❖ Allows staying in touch while on the move.
- ❖ Useful for emergency situations without public phones.
- ❖ Facilitates business and personal calls anywhere.
- ❖ Mobile networks are less stable than landlines.
- ❖ Signal quality depends on network coverage.
- ❖ Essential for keeping contact with co-workers remotely.

Internet Telephony

- ➢ Method for talking to people using the internet.
- ➢ Converts sound into digital data packets for transmission.
- ➢ Accessed via mobile networks or broadband.
- ➢ Calls are often free regardless of global location.
- ➢ Issues include sound quality, echo, and weird sounds.
- ➢ Security risks include identity theft and malware.

Video Calling

- ❖ Uses built-in cameras to add video to VoIP calls.
- ❖ Cheaper than traditional video conferencing systems.
- ❖ Apps like FaceTime and Zoom are common examples.
- ❖ Split screens allow seeing multiple people simultaneously.
- ❖ Zoom allows recording sessions for later playback.
- ❖ Cloud-based services reduce infrastructure costs.

6.2

Modelling Applications

Spreadsheet Models

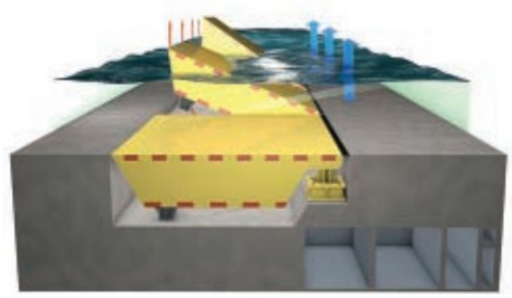
- ❖ Spreadsheets calculate profit and loss based on variables.
- ❖ "What-if" scenarios allow testing different financial outcomes.
- ❖ Mathematical formulas represent real-world financial situations.
- ❖ Used to forecast business performance over time.
- ❖ Variables like costs and revenue can be adjusted easily.
- ❖ Essential for budgeting and strategic planning.

Civil Engineering Modelling

- ❖ Engineers use 3D computer modelling for bridges and buildings.
- ❖ Designs are tested long before construction begins.
- ❖ Simulations check for structural integrity under stress.
- ❖ Tests include wind tunnel effects and earthquake resilience.
- ❖ Allows zooming and rotation to view fine details.
- ❖ Identifies design flaws early to save money and lives.

Flood Water Management

- ❖ Used to predict water levels and potential flood depths.
- ❖ Inputs include river cross-sections and bridge dimensions.
- ❖ Factors like tides and upstream flows are considered.
- ❖ Simulations help plan flood defences and mitigation.
- ❖ Past flooding data calibrates the model for accuracy.
- ❖ Models update continuously as defences are built.



▲ **Figure 6.9** Flood management barriers

Venice Example

- Venice uses models to manage flood barriers.
- Simulations predict barrier reactions to flood scenarios.
- Helps determine when to raise barriers based on tides.
- Prevents damage from rising sea levels.
- Complex calculations are handled by computer systems.
- Ensures the safety of the city's infrastructure.

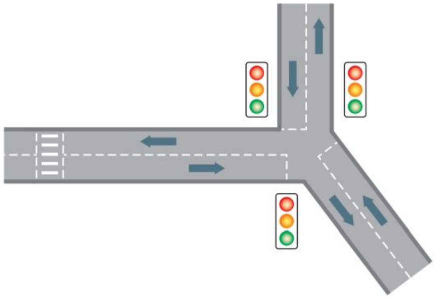
Traffic Simulations

- ❖ Models traffic flow during roadworks or accidents.
- ❖ Tests the effect of different speed limits on flow.
- ❖ Helps design road layouts to minimize congestion.
- ❖ Simulates breakdown scenarios to assess impact.
- ❖ Optimization allows for faster repairs and better flow.
- ❖ Cheaper and safer than physical road closures for testing.

Traffic Light Simulation

Controlling Junctions

- ❖ Models traffic lights at complex junctions (e.g., Y-junctions).
- ❖ Data is collected via induction loop sensors.
- ❖ Records vehicle counts, timing, and build-up at lights.
- ❖ Simulations test light timing variations for efficiency.
- ❖ Considers emergency vehicles and pedestrian crossings.
- ❖ Real-time systems adjust lights based on current traffic.



▲ **Figure 6.11** Traffic light simulation

Traffic Simulation Data Collection

- Counting the number of vehicles passing through the junction from all possible directions.
- Recording the specific time of day alongside vehicle counts to identify peak hours.
- Measuring how many vehicles build up (queue) at the junction during different periods.
- Collecting data across weekends and holidays to account for changing traffic patterns.
- Timing how long it takes the fastest and slowest vehicles to clear the junction.
- Accounting for nearby pedestrian crossings, filtering lanes, and specific turn types (left vs. right turns).

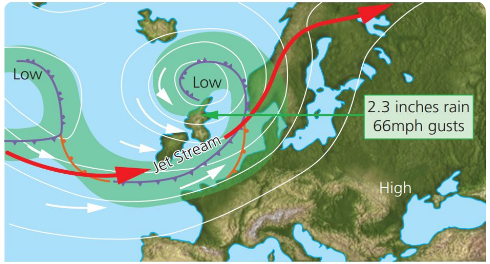
- **Model Validation:** Designers first run the model and compare its results with real-world traffic data to ensure accuracy.
- **What-if Scenarios:** Once validated, designers vary the timing of the lights to observe the impact on traffic flow.
- **Dynamic Testing:** Adjusting the number of stopped vehicles or changing traffic volumes in all directions to find the "breaking point."
- **Emergency Priority:** Simulating how emergency vehicles (ambulances or fire trucks) affect the flow at different times of the day.

Weather Forecasting

- ❖ Weather stations use specialized sensors to automatically gather continuous data on rainfall, temperature, wind, and humidity.
- ❖ Environmental data is gathered every hour of every day to ensure the simulation has a constant stream of real-time information.
- ❖ The atmosphere is divided into a three-dimensional grid where complex mathematical equations represent the exchange of heat and moisture.
- ❖ Forecasters run data through the computer model to simulate air movement and predict weather patterns for several days in advance.

Weather Forecasting

- ❖ The model "learns" by comparing its previous forecasts against actual weather outcomes to improve the accuracy of future predictions.
- ❖ Powerful computers are utilized to perform "number crunching" on vast amounts of data required for complex global simulations.
- ❖ Results are often converted into animated formats, projecting weather symbols onto maps to show real-time changes to the public.



▲ **Figure 6.12** Animated weather forecast

6.3

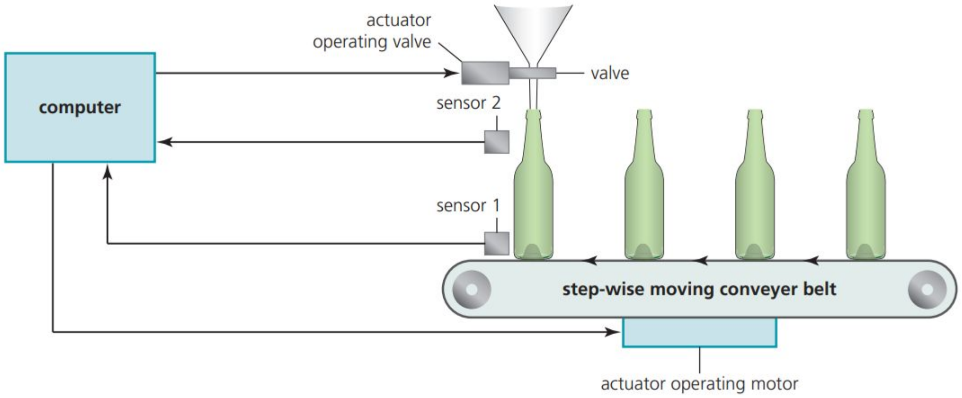
Computer controlled systems

Role of Sensors and Actuators

- ❖ Computers use sensors to monitor physical environments.
- ❖ Data is compared against preset values or rules.
- ❖ Decisions trigger actuators to perform tasks.
- ❖ Used in manufacturing, robotics, and automation.
- ❖ Operates continuously without human intervention.
- ❖ Ensures precision and consistency in tasks.

Bottling Plant

- ❖ Robots handle tasks like filling, capping, and moving bottles.
- ❖ Sensors detect bottle presence and liquid levels.
- ❖ Actuators open/close valves and move conveyor belts.
- ❖ Process is continuous until stopped for maintenance.
- ❖ Ensures consistent filling and production speed.
- ❖ Reduces the need for manual handling of goods.



Benefits of Automation

- Higher productivity and consistency than human workers.
- Can work 24/7 without breaks or holidays.
- Operates safely in hazardous environments (e.g., chemicals).
- Reduces long-term wage costs for the company.
- Greater precision minimizes waste and errors.
- Boredom from repetitive tasks is eliminated.

Drawbacks of Automation

- High initial setup and programming costs.
- Leads to unemployment among manual workers.
- Robots lack the ability to think or improvise.
- Maintenance requires expensive, specialized staff.
- Errors in programming can lead to massive production faults.
- Security risks if the system is hacked.

6.4

School Management Systems

Student Registration

- Tracks daily attendance of students.
- Methods include OMR, magnetic stripes, and biometrics.
- Electronic systems are faster than paper calls.
- Data is stored centrally for easy reporting.
- Helps identify truancy patterns quickly.
- Automated messages can be sent to parents.

Registration Methods

- **OMR**: Scans paper registers with pencil marks.
- **Smart Cards**: Students swipe cards upon entry.
  - Can also be used for cashless catering.
- **Biometrics**: Fingerprints used for identification.
  - Can prevent students from signing in for others.
- GPS tracking is a potential (but controversial) future method.

Academic Monitoring

- ❖ Spreadsheets record test results over the year.
- ❖ Teachers compare performance against class averages.
- ❖ Data is easily imported into report cards.
- ❖ Tracks both academic achievement and behaviour.
- ❖ Summaries can be generated for parents or administrators.
- ❖ Identifying underperformance allows for early intervention.

6.5

Booking Systems

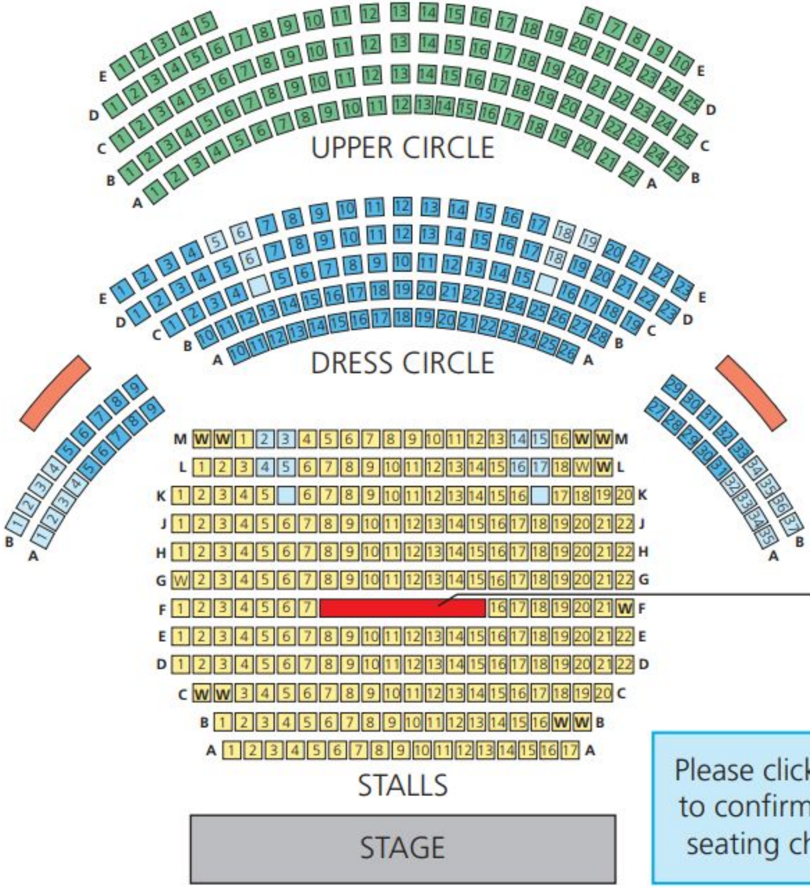
Travel Industry

- ❖ Used for booking flights, holidays, and hotels.
- ❖ Customers search for flights by destination and date.
- ❖ Comparison of prices and airlines is simplified.
- ❖ Home printing of boarding passes is possible.
- ❖ Prevents double-booking through real-time databases.
- ❖ Accessible 24/7 from anywhere with internet.

Booking Entertainment

Theatres and Cinemas

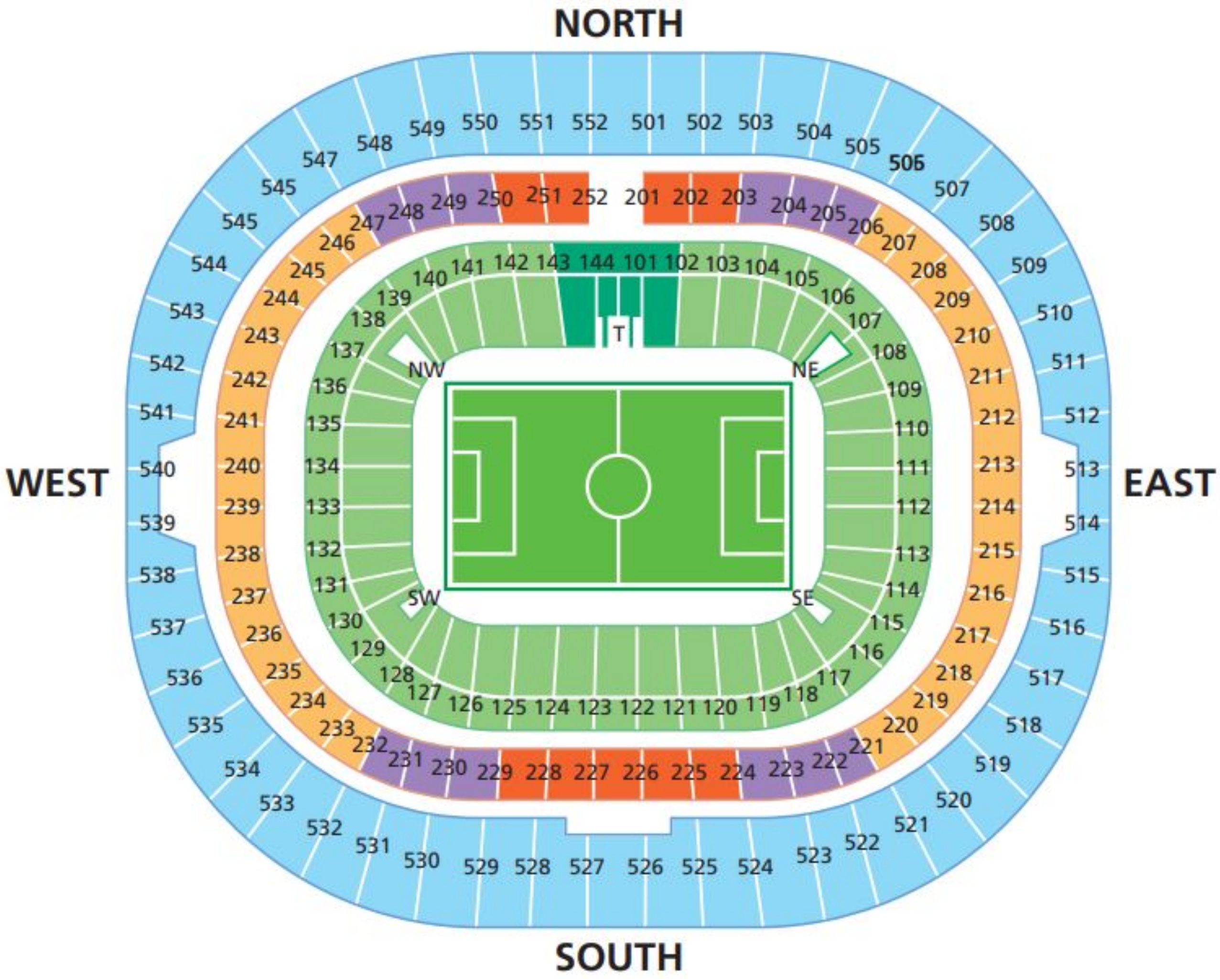
- ❖ Users view real-time seating plans.
- ❖ Allows selection of specific seats and prices.
- ❖ E-tickets are sent via email or QR code.
- ❖ QR codes are scanned at the venue entrance.
- ❖ "Impulse" bookings can be made shortly before events.
- ❖ Special offers can be targeted based on booking history.



The seating display at the venue is shown on the screen

The user selects their seat(s) by highlighting the actual seats on the screen display and then clicks CONFIRM to go to the next part of the process

Please click here to confirm your seating choice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Advantages of Online Booking

- ❖ Convenience of booking from home at any time.
- ❖ Easy comparison of prices and options.
- ❖ Immediate confirmation and e-ticket delivery.
- ❖ Reduces staffing costs for booking offices.
- ❖ Global audience reach for events and travel.
- ❖ Special offers can be easily distributed.

Disadvantages of Online Booking

- Requires computer/mobile and internet access.
- Harder to cancel or get refunds compared to person-to-person.
- Server crashes can prevent bookings entirely.
- Phishing sites can steal credit card details.
- Lack of personal interaction for complex queries.
- System maintenance is expensive.

6.6

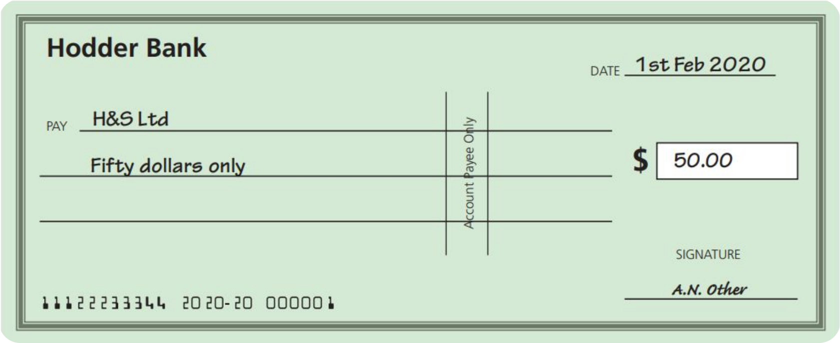
Banking Systems

Banking Applications

- ❖ Includes ATMs, online banking, and cheque processing.
- ❖ ATMs allow cash withdrawal and bill payments 24/7.
- ❖ Online banking manages accounts from home.
- ❖ Chip and PIN technology secures card transactions.
- ❖ Electronic funds transfer moves money instantly.
- ❖ Technology has reduced the need for physical branches.

Cheque Processing

- Cheques are scanned to create a digital image.
- OCR reads the sort code and account number.
- Magnetic Ink Character Recognition (MICR) is used for accuracy.
- "Out clearing" and "in clearing" manage the transfer.
- Funds are transferred electronically between banks.
- Digital imaging speeds up the clearing cycle significantly.



Using Cheques

Advantages	Disadvantages
» more convenient and safer than cash » it is possible to stop payments if necessary » a cheque can be drawn any time (up to six months after it was dated and signed) » cheques can be post-dated » cheques can be traced if they are 'lost'	» cheques are not legal tender and can be refused » it is a slow method of payment » easier for fraudsters than credit card or debit card payment methods » relatively expensive payment method

Chip and PIN Cards

- ❖ Microchip stores critical account data securely.
- ❖ Users enter a 4-digit PIN to authorize payments.
- ❖ PIN must match the data stored on the chip.
- ❖ More secure than magnetic stripe signatures.
- ❖ Reduces fraud from lost or stolen cards.
- ❖ Readers do not always need a live connection to authorize.

Contactless and NFC

- ❖ Uses Near Field Communication (NFC) technology.
- ❖ Cards or phones are held close to the reader (< 5cm).
- ❖ Transactions are encrypted and dynamic for security.
- ❖ Ideal for small, low-value purchases.
- ❖ Speeds up checkout times significantly.
- ❖ Mobile wallets (e.g., Apple Pay) use this technology.

Type of card	Advantages	Disadvantages
Credit	» there is customer protection if a company stops trading or goods do not arrive » internationally accepted method of payment » interest-free loan if money paid back within agreed time period » can buy items online	» can be charged high interest rate » annual fees often apply » easy to end up with credit damage as sums mount up » security risks when using credit card online (see Chapter 8)
Debit	» money comes from customer's current account, therefore no interest charges » safer than carrying cash » can buy items online	» less customer protection than credit card if goods do not arrive or company goes out of business » no credit allowed; customers must have the funds available » security risks when using debit card online (see Chapter 8)

Advantages of ATMs

- ❖ Can withdraw cash at any time of day.
- ❖ Offers banking services without the need to go into the bank.
- ❖ Access an account from anywhere in the world.
- ❖ Provides quicker service than waiting in a queue in a bank.

Disadvantages of ATMs

- ❖ Often in places where theft can take place at night.
- ❖ Potential for shoulder-surfing and card-cloning scams.
- ❖ Banks charge customers for using ATMs.
- ❖ Withdrawal limits are often imposed on customers.
- ❖ If the debit card is faulty then no transaction can take place.
- ❖ Loss of the personal touch (Face to face interactions).

6.7

Computers in Medicine

Information Systems in Medicine

- ❖ Patient records are stored in large databases.
- ❖ Accessible by doctors and pharmacists instantly.
- ❖ Ensures correct diagnosis and avoids drug interactions.
- ❖ Critical for accessing history in emergencies.
- ❖ Digital prescriptions reduce paper usage.
- ❖ Unique IDs track patients across the system.

3D Printing in Medicine

- ❖ Used in Medical Manufacturing
- ❖ Creates prosthetics and artificial limbs.
- ❖ Produces complex tissue structures and blood vessels.
- ❖ Customized medicines can be printed for patients.
- ❖ Surgical tools can be made on-demand.
- ❖ Reconstructive surgery uses printed bone structures.
- ❖ Revolutionizing personalized medical care.

6.8

Expert Systems

Usage

❖ Simulates the knowledge of a human expert.

❖ Used for diagnosis, prospecting, and planning.

❖ Solves complex problems by reasoning with data.

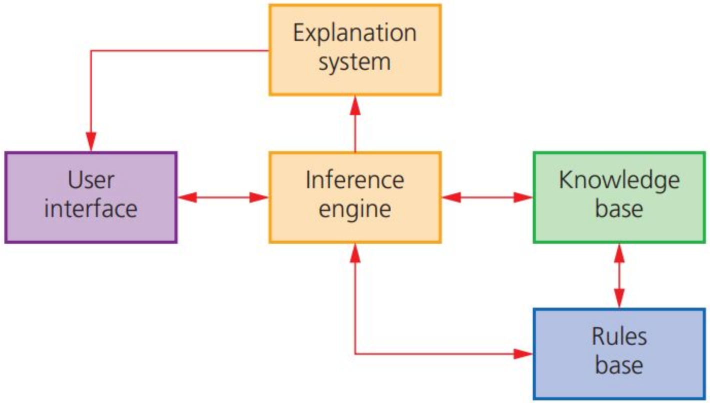
❖ Can operate 24/7 without fatigue.

❖ Provides consistent answers based on logic.

❖ Preserves expertise even if humans leave.

Components of an Expert System

- **User Interface:** Allows interaction via questions/prompts.
- **Knowledge Base:** Database of facts and rules.
- **Inference Engine:** Applies rules to the knowledge base.
- **Rules Base:** Logic (If-Then statements) driving decisions.
- **Explanation System:** Explains reasoning to the user.
- All components work together to provide a solution.



How Expert Systems Work

1. User enters data via the user interface.
2. Inference engine searches the knowledge base.
3. Rules are applied to find matches or solutions.
4. System calculates the probability of accuracy.
5. Conclusion and explanation are presented to the user.

- Can suggest further actions or remedies.

Medical Diagnosis

- ❖ System asks questions about symptoms.
- ❖ User answers yes/no or multiple-choice questions.
- ❖ Inference engine matches symptoms to known illnesses.
- ❖ Suggests diagnosis and treatment with probability.
- ❖ Explanation system justifies the result.
- ❖ Helps doctors narrow down possibilities quickly.

Example 1: medical diagnosis

Input screen

- First of all an interactive screen is presented to the user
- The system asks a series of questions about the patient's illness
- The user answers the questions asked (either as multiple-choice or yes/no questions)
- A series of questions are asked based on the user's responses to previous questions

Expert system

- The inference engine compares the symptoms entered with those in the knowledge base looking for matches
- The rules base is used in the matching process
- Once a match is found, the system suggests the probability of the patient's illness being identified accurately
- The expert system also suggests possible solutions and remedies to cure the patient or recommendations on what to do next
- The explanation system will give reasons for its diagnosis so that the user can determine the validity of the diagnosis or suggested treatment

Output screen

- The diagnosis can be in the form of text or it may show images of the human anatomy to indicate where the problem may be
- The user can request further information from the expert system to narrow down the possible illness and its treatment

Oil Prospecting

- Users input geological data and profiles.
- System analyzes data against known geological models.
- Predicts the probability of finding oil deposits.
- Suggests the depth of deposits and drilling locations.
- Produces contour maps of mineral concentrations.
- Reduces the cost of exploratory drilling.

Route Scheduling

- Determines efficient routes for delivery vehicles.
- Considers distance, traffic, and number of drops.
- Calculates the least expensive and fastest route.
- Suggests the optimal number of vehicles and drivers.
- Used by courier and shipping companies.
- Saves fuel and time in distribution.

6.9

Computers in the retail industry

Point of Sale (POS)

- Terminals replace traditional cash registers.
- Barcodes are scanned to identify products.
- Prices are retrieved from a central database.
- Itemized receipts are printed for customers.
- Stock levels are automatically updated.
- Allows for better inventory management.

Barcodes

- ❖ Series of black and white lines representing numbers.
- ❖ Scanned by laser at the checkout.
- ❖ Identifies the country, manufacturer, and product code.
- ❖ Does not contain the price (price is in the database).
- ❖ Allows quick price changes without re-tagging items.
- ❖ Speeds up the checkout process for customers.

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Electronic Funds Transfer (EFTPOS)

- ❖ Terminals accept credit and debit cards.
- ❖ Checks card validity and available funds.
- ❖ Transfers money from customer to shop account.
- ❖ Secure communication with the bank is essential.
- ❖ Provides a record of the transaction.
- ❖ Reduces the amount of cash held in shops.

Online Shopping & E-commerce

- ❖ Buying goods and services via the internet.
- ❖ Customers browse catalogues and add to baskets.
- ❖ Secure checkout handles payment and delivery details.
- ❖ Goods are delivered directly to the customer.
- ❖ Comparison sites help find the best deals.
- ❖ Access to global markets for niche products.

Advantages of Online Shopping

- ❖ Shop 24/7 from the comfort of home.
- ❖ Easy to compare prices between different vendors.
- ❖ No travel costs or parking fees involved.
- ❖ Access to a wider variety of goods.
- ❖ Disabled or elderly people can shop easily.
- ❖ Read reviews from other customers before buying.

Disadvantages of Online Shopping

- ❖ Cannot physically inspect or try on items.
- ❖ Delay between purchase and delivery.
- ❖ Risk of credit card fraud and fake websites.
- ❖ Returning goods can be costly and time-consuming.
- ❖ Lack of social interaction found in high streets.
- ❖ Shipping costs can add to the total price.

Social and Economic Effects of Online Shopping

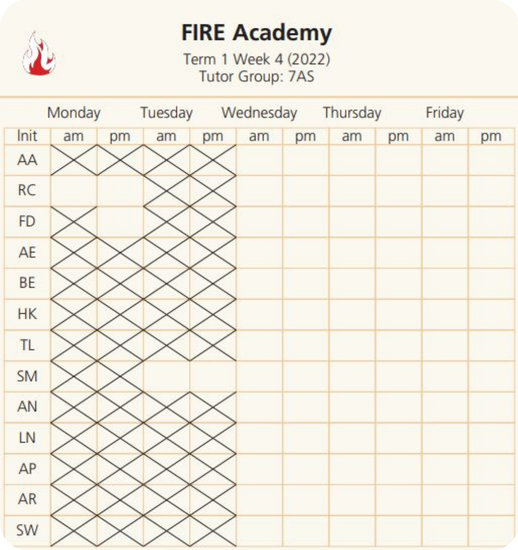
- High street shops are closing due to competition.
- Increased delivery traffic causes pollution.
- "Ghost towns" created by deserted shopping areas.
- Job losses in retail but gains in logistics.
- Health risks associated with lack of movement.
- Digital divide affects those without internet access.

6.10

Recognition systems

Optical Mark Recognition (OMR)

- ❖ Reads pencil/pen marks on a pre-defined form.
- ❖ Used for lottery tickets, exams, and registers.
- ❖ Sensors detect the position of marks (lozenges).
- ❖ Fast and accurate for processing large data volumes.
- ❖ Coordinates of marks translate to data values.
- ❖ Requires forms to be filled in correctly.


FIRE Academy
Term 1 Week 4 (2022)
Tutor Group: 7AS

	Monday		Tuesday		Wednesday		Thursday		Friday	
Init	am	pm	am	pm	am	pm	am	pm	am	pm
AA	X	X	X	X						
RC			X	X						
FD	X		X	X						
AE	X	X	X	X						
BE	X	X	X	X						
HK	X	X	X	X						
TL	X	X	X	X						
SM	X	X								
AN	X	X	X	X						
LN	X	X	X	X						
AP	X	X	X	X						
AR	X	X	X	X						
SW	X	X	X	X						

OMR in Assessments

- ❖ Completed forms scanned by OMR readers.
- ❖ Timing marks align the paper for reading.
- ❖ Answers are compared to a stored answer key.
- ❖ Grading is instant and free of human bias.
- ❖ Results are stored in a database for analysis.
- ❖ Cannot read handwriting, only marks.

Optical Character Recognition (OCR)

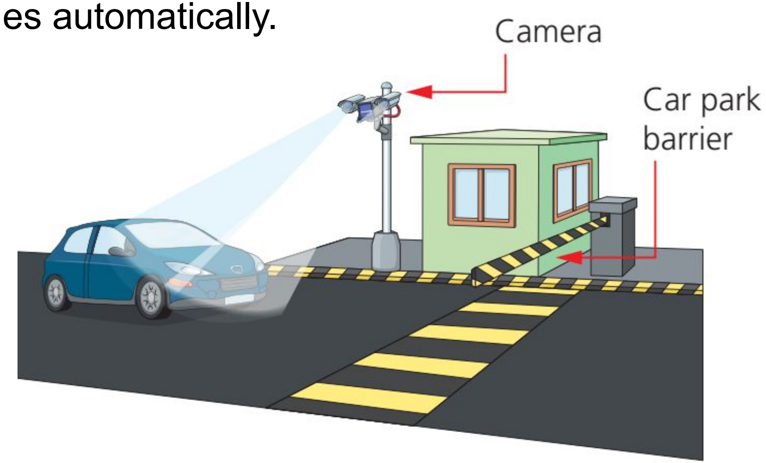
- Converts scanned images of text into editable text.
- Used for digitizing books and documents.
- Allows searching and editing of scanned content.
- Used in passport reading and cheque processing.
- Software compares shapes to known character sets.
- Accuracy depends on the quality of the original.

Automatic Number Plate Recognition (ANPR)

- ❖ Cameras capture images of vehicle number plates.
- ❖ OCR software converts image to text numbers.
- ❖ Used for car park entry/exit and toll collection.
- ❖ Detects speeding and stolen vehicles automatically.
- ❖ Calculates travel times for traffic management.
- ❖ Privacy concerns exist regarding surveillance.

ANPR Applications

- ❖ Controls access to private car parks.
- ❖ Enforces congestion charges in cities.
- ❖ Alerts police to vehicles of interest.
- ❖ Monitors average speed over road sections.
- ❖ Provides data for transport planning.
- ❖ Can be used to issue parking fines automatically.

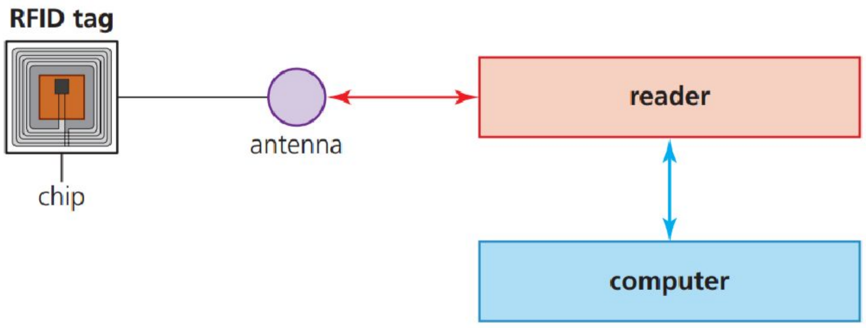


Quick Response Codes

- ❖ 2D matrix barcodes holding more data than standard barcodes.
- ❖ Read by smartphone cameras and apps.
- ❖ Links to websites, videos, or contact info.
- ❖ Used in advertising, tickets, and product info.
- ❖ No need for expensive laser scanners.
- ❖ Can be printed on any medium easily.

Radio Frequency Identification

- ❖ Uses radio waves to read data from tags.
- ❖ Tags consist of a microchip and antenna.
- ❖ Can be read from a distance without line-of-sight.
- ❖ Used for inventory tracking and animal tagging.
- ❖ Bulk reading allows multiple items to be scanned at once.
- ❖ More durable than barcodes.



▲ **Figure 6.34** RFID tag showing antenna and chip

RFID in Passports

- ❖ RFID chips embedded in passport covers.
- ❖ Stores biometric data (face, fingerprints) and ID details.
- ❖ Scanners power the passive chip to read data.
- ❖ Enhances security and prevents forgery.
- ❖ Speeds up immigration control via e-gates.
- ❖ Data is encrypted to protect privacy.

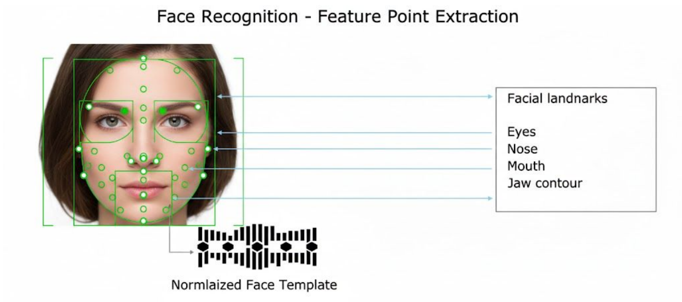
Biometric Recognition

- ❖ Uses unique biological characteristics.
- ❖ Common types: Fingerprint, Face, Iris, Voice.
- ❖ Used for security access and device unlocking.
- ❖ Data is captured, digitized, and compared to records.
- ❖ Difficult to forge or fake.
- ❖ Raises privacy and data storage concerns.

Fingerprint and Face Recognition

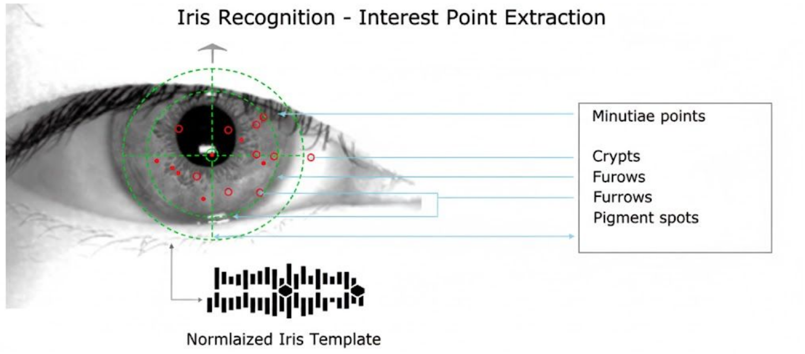
- ❖ Fingerprint: Scans unique ridge patterns. Used on phones/laptops.
- ❖ Fingerprint is widely accepted and cheap to implement.
- ❖ Cuts on fingers can cause reading errors.

- ❖ Face: Analyzes facial features and distances.
- ❖ Face recognition used in airports and policing.
- ❖ Face recognition can struggle with lighting or glasses.



Iris Recognition

- Scans the unique patterns in the colored iris.
- Uses visible and near-infrared light cameras.
- Very high accuracy and difficult to spoof.
- Used in high-security areas and border control.
- Fast verification process (under 5 seconds).
- Works even with contact lenses or glasses.

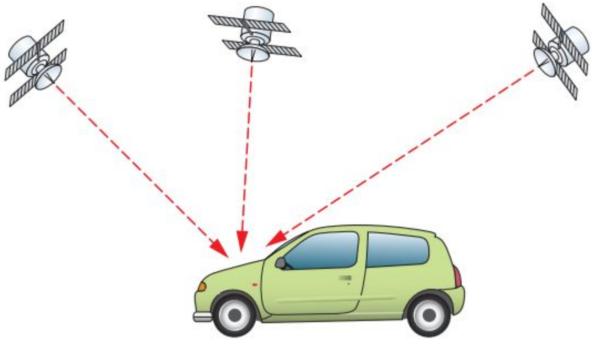


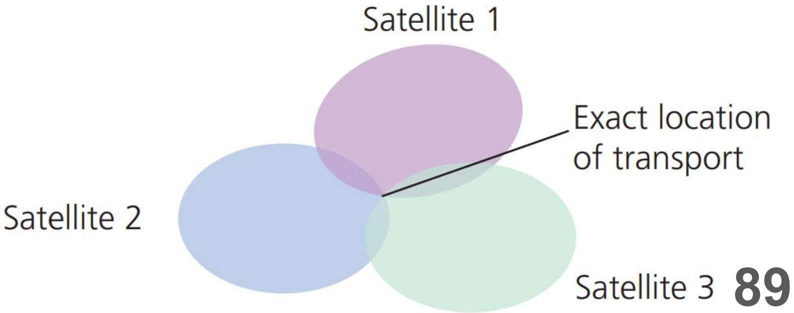
6.11

Satellite systems

Global Positioning System

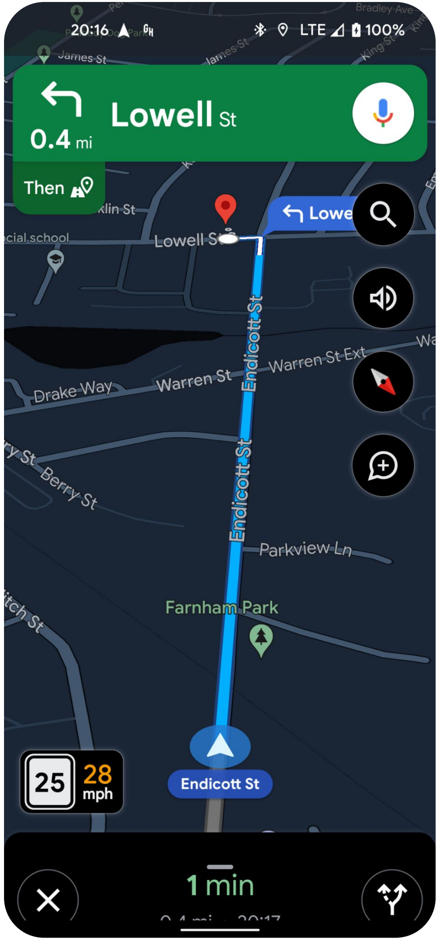
- Determines exact location using satellites.
- Satellites transmit position and time data.
- Receivers calculate location using triangulation.
- Requires signals from at least three satellites.
- Used in cars (Satnav), ships, and planes.
- Atomic clocks ensure extreme timing accuracy.





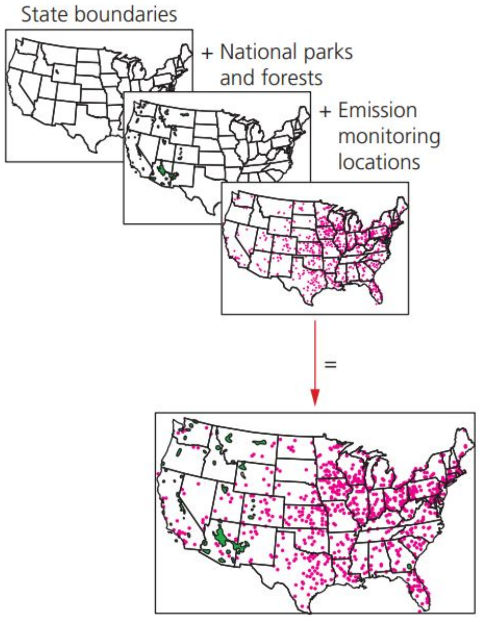
Navigation and Tracking

- Provides turn-by-turn directions for drivers.
- Hikers and runners track routes and distances.
- Farmers map fields for precision agriculture.
- Geocaching games use GPS coordinates.
- Emergency services locate distress signals.
- Allows fleet management for trucking companies.



Geographic Information Systems

- ❖ Captures, stores, and analyzes geographic data.
- ❖ Layers data (e.g., roads, rivers, population) on maps.
- ❖ Used for urban planning and environmental studies.
- ❖ Helps decide locations for new schools or roads.
- ❖ Combines database data with map visualizations.
- ❖ Essential for analyzing spatial relationships.



Satellite Media Communication

- Transmits TV, phone, and internet signals.
- Signals sent from Earth to satellite and back.
- Covers large areas and remote terrain.
- Cheaper than laying cables over long distances.
- High bandwidth supports media broadcasting.
- Signal delay (latency) can occur due to distance.

Satellites

Artificial satellites are specialized man-made spacecraft launched into orbit around Earth and other planets in the solar system. They are vital in telecommunications because they can receive signals from the ground and amplify and retransmit them to distant parts of Earth.

Communications satellites

Communications satellites are designed to send and receive radio signals carrying audio, video, and other kinds of data. Relaying signals via satellite allows for rapid communication across huge distances. Signals of set frequencies are sent into space from ground stations and picked up by a satellite's antenna. A transponder processes the information and boosts the signal before relaying it down to other ground stations on Earth.

SPUTNIK 1 WAS THE FIRST SATELLITE IN SPACE, LAUNCHED BY THE SOVIET UNION ON OCTOBER 4, 1957

Uses of satellites

Although the first satellites were launched during the Cold War for the purposes of space exploration and defense, they are now engineered for a wide range of military and civilian uses. Most people make use of satellites every day without even realizing it.

Astronomical
Satellite-based telescopes are ideal for observing space because, unlike terrestrial telescopes, they are not obstructed by Earth's atmosphere.

Telephones
Satellite phones exchange signals with satellites instead of ground-based cell towers. They are often used in remote areas not covered by terrestrial signals.

Television
Many television broadcasters deliver programs via satellite. Viewers receive the signals using a satellite dish attached to the outside of their home.

Radio
Relaying radio programs via a satellite means that signals can be transmitted across entire countries.

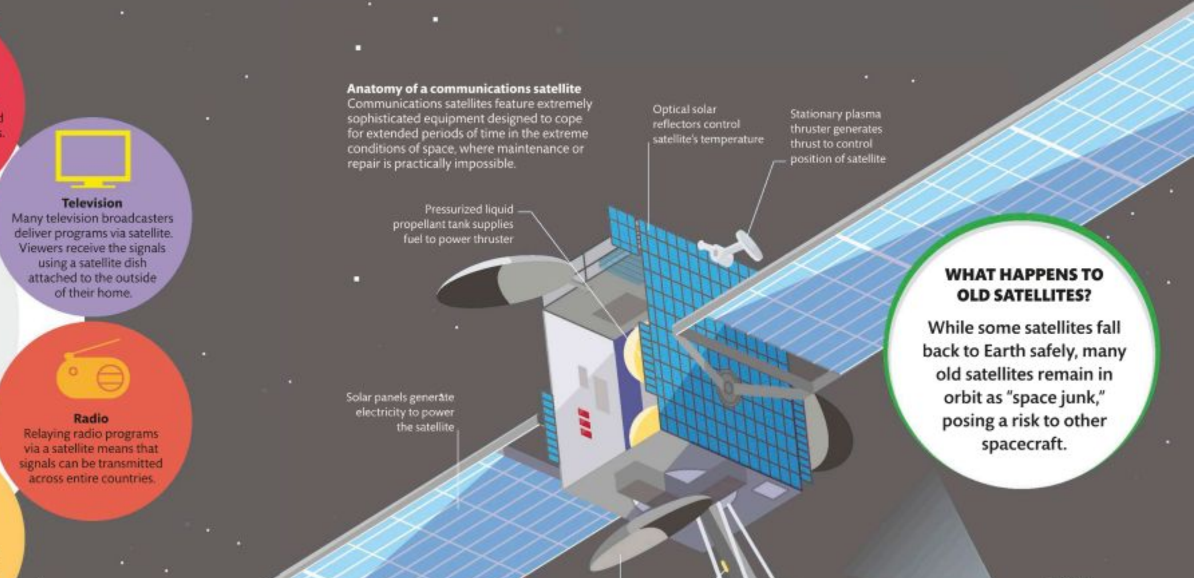
Internet
Satellite Internet provides coverage in remote areas, but service can be slow due to the distance signals have to travel.

GPS navigation
Navigation devices can determine their position on Earth by exchanging information with satellites (see pp.194–195).

Weather
Some satellites are designed to monitor features of Earth's weather and climate. They transmit the data back to Earth for analysis.

Defense
Military satellites have a range of uses, including surveillance, navigation, and for sending encrypted communications.

Anatomy of a communications satellite
Communications satellites feature extremely sophisticated equipment designed to cope for extended periods of time in the extreme conditions of space, where maintenance or repair is practically impossible.

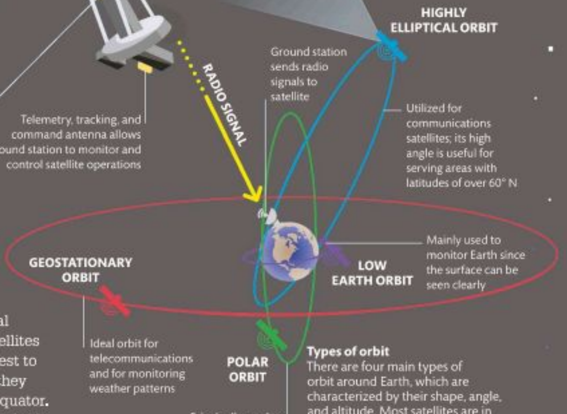


WHAT HAPPENS TO OLD SATELLITES?

While some satellites fall back to Earth safely, many old satellites remain in orbit as "space junk," posing a risk to other spacecraft.

Satellite orbits

Satellites enter orbit if they launch at high enough velocity to overcome Earth's gravitational pull at the surface and then balance its weaker gravitational pull in space. Many communications satellites are in geostationary orbit. They travel west to east at the same rate as Earth spins, so they appear stationary above a point on the equator. Some satellites have polar orbits, crossing both poles on their journey around Earth.



Types of orbit
There are four main types of orbit around Earth, which are characterized by their shape, angle, and altitude. Most satellites are in low earth orbit, less than 1,250 miles (2,000 km) above the surface.